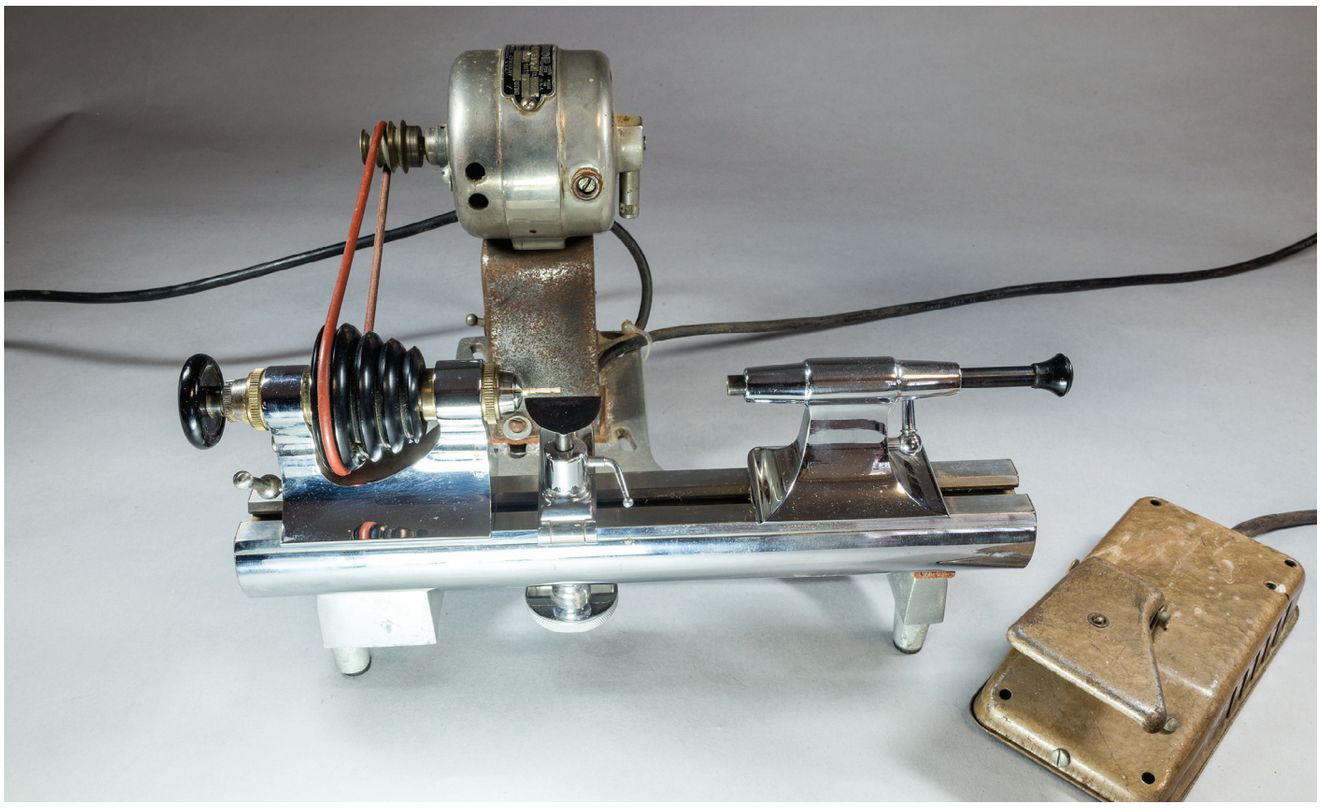
Lathes

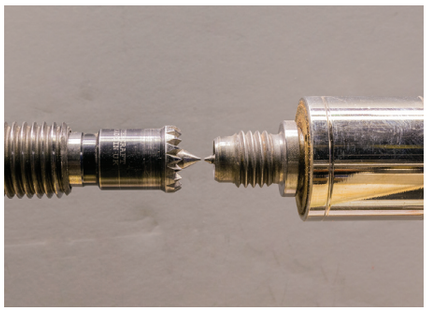
Regardless of the size of lathe that you own, you can do smaller and miniature work provided that you can hold the material and the lathe runs true enough for the size of work you plan. Obviously, a 660mm swing might be overkill but it certainly will work. Whether you turn on a full-sized lathe or a jewellers lathe, running true is key as is the ability to get in close with a 'right' sized tool rest. You may not be using the tailcentre for your projects. Many will be done as faceplate-type holding alone. If so, the alignment of the tailcentre with the headstock will not be a critical point for these turnings. You'll turn with respect to the headstock and all will be created on that axis of rotation.

If you do need to incorporate the tailcentre for your work, it is key that the headstock and tailstock alignment is as good as can be. Any alignment errors can only aggravate your efforts for precision work. This simple alignment check, and adjustment if needed, is quickly and easily done and is important to perform regardless of what you are turning. Once done, it probably will never need to be done again, although checking on occasion is a wise idea. No special tools are needed. You can perform the alignment check with two spur centres that have a sharp point inserted. Be certain that the Morse tapers in both the headstock and tailstock are clean prior to inserting the drive centres in each. The centres need only be seated firmly by hand. There's no need to drive them into place with a lot of force. If they are seated in the clean taper, they'll work fine for this check. You can also use a couple of dead centres or a drive centre and tail centre with a pin. Slide the tailstock forward until the tips of the two points just barely touch. Examine the alignment from the front and top along with the axis of both. If there is a mismatch, you can check your lathe for methods to correct for misalignment. Many have a set of bolts that mount the headstock to the lathe bed or ways. Depending on what type of correction is needed, loosen the bolts and shim the offending areas with tape, aluminum foil, playing card stock, or actual metal shim stock.

This is a trial and error type of adjustment because once the bolts are tightened again, you may need to revisit the shimming operation. Repeat until you've achieved the best you can. This entire process is covered in more detail in *Woodturning* 254 if you'd like to find more information on the technique.



Lathes come from pretty big to pretty small. This fully functional variable speed jewellers lathe can be carried in one hand



Anything with a sharp point can be used to check headstock to tailstock alignment. Take the time to check and adjust for error

Whether bolted from the top or bottom, nearly all headstock mounting holes have some room for minor alignment adjustment

When shimming is required for alignment, playing cards, tape, or a proper amount of folded aluminium foil will work well

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